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Fairphone 2 Reviewed

The much talked about, user-repairable, scoring a 10 on 10 iFixit score, the Fairphone 2 Review is now out! Have you read it yet? <http://dgit.in/FairPhone2Rw>

fifth-generation air superiority platform by far happens to be the the stealth jet arms race kickstarted by China with its own fifth-generation stealth fighter in the last decade. The stealth fighter took everyone by surprise because until then China was considered incapable of making competitive fourth or even third generation aircraft, let alone possess the expertise and know-how to develop a sophisticated fifth-generation air dominance fighter.

Thanks to Edward Snowden's efforts, it was later revealed that China had essentially stolen several terabytes of critical F-35 data – on everything from radar systems, engine schematics, and heat-masking design implementations to super secret information on stealth systems – right off Lockheed Martin's servers. What's more, the Chinese had also managed to swipe F-22 blueprints to create its own copycat versions of USAF's F-22 and F-35 stealth aircraft. The more sophisticated Chengdu J-20 Black Eagle is the equivalent of the F-22 Raptor, whereas the Shenyang J-31 Gyrfalcon is nearly the exact copy of the F-35 Lightning II.



The ShinShin in all its glory

If the capabilities of the ShinShin stealth fighter are to be compared, it must be pit against its most likely regional foes – the J-20 and (to a lesser extent) J-31. However, unlike the Chinese stealth fighter jets slated to enter service within the next two years, the Japanese Shinshin isn't a final design and won't be deployed anytime before 2030. Mitsubishi is still in the R&D phase, and the specifications on the ShinShin research concept are essentially projections that are subject to change pending extensive

feasibility tests. In the simplest of terms, such a comparison would be akin to pitting the existing capabilities of the Chinese stealth jets against what the Japanese hope to incorporate into their F-3 stealth platform over the course of the next decade.

How does it compare?

While the information on the J-20 is sketchy, it's capable of supercruise. In other words, it can achieve supersonic flight without employing fuel-inefficient afterburners. The Russian Sukhoi PAK FA achieves the same feat at Mach 1.6, whereas the F-22 is a bit faster at Mach 1.82. The J-20, however, has a relatively slower maximum speed of Mach 2.0, as opposed to the F-22's Mach 2.25 and the PAK FA's Mach 2.3. The ShinShin aims to achieve a max speed exceeding Mach 2 with its 11,000 lbf twin turbofan engines, with a supercruise capability on par with that of the F-22. Like its contemporaries, the Japanese stealth aircraft will incorporate advanced three-axis 3D thrust vectoring technology similar to that seen on the US Rockwell X-31 experimental aircraft. If Mitsubishi were to pull this engine off, it would enable the Shinshin to achieve a crazy angle of attack and perform spectacular feats of agility that should come handy in dogfights.

In its current X-2 iteration, the ShinShin has a projected combat range of 2,900 km and a combat range of 761 km, which is on par with the F-22, but only a fraction of the range achievable by the PAK FA, which can reach out to a maximum of 3,500 km. This doesn't really matter much, since Japan is essentially concerned with defending its own airspace or reaching North Korea, which falls well within range. What's more interesting is the upgrade to avionics, which surpasses the current standards for fifth-generation aircraft. Copper-based fly-by-wire controls are being replaced by faster and interfer-



The X-2 aims to reach speeds in excess of Mach 2

ence resistant optical fibre connections. In addition to advanced AESA radars, the ShinShin's avionics will incorporate sophisticated survivability implementation dubbed self repairing flight control capability that allows the aircraft to compensate for damaged control surfaces and dynamically reconfigure working ones to regain control.

A bird in hand...

Any further comparisons are moot since not much else is known about the Mitsubishi ATD-X platform. Furthermore, all of these specifications are goals that will invariably be subject to revisions, additions, and even omissions pending the many flight tests and redesigns that will take place over the course of the next decade. Comparing the ShinShin to existing fifth-generation offerings from China, Russia, and USA is a bit like comparing apples to non-existent oranges that haven't even been harvested yet.

However, it is worthwhile to note that the ShinShin project initially began as a sort of Japanese bargaining chip to pressure USA into selling its F-22 fighters after the Chinese J-20 stealth aircraft was revealed. The programme only assumed any degree of seriousness near the end of the last decade when US Congress made it clear that it wouldn't be selling the F-22 to anyone. The rumoured involvement of F-22 creator Lockheed Martin and the ShinShin's cosmetic and operational similarities to the US air superiority fighter could very well translate into a force to be reckoned with. Just remember that it's all conjecture at this stage, when the cold, hard reality is that the Chinese and the Russians will have inducted operational units in the next two years. You can't possibly beat that with an unproven concept. 